

Graph Retrieval-Augmented Language Model for Question Answering of Vietnamese Law

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Abstract. Nowadays Large Language Models (LLM) have been used for basic tasks such as question answering and information retrieval. However, these models struggle to adapt to deep, high-precision professional knowledge domains, such as healthcare and law, without relying on costly fine-tuning methods when applied to non-English languages without large-scale training corpora. This paper focuses on the Vietnamese legal domain, where legal knowledge is highly structured but complex, with intricate relationships, and large-scale annotated data remain limited. The legal information retrieval system for Vietnamese law is designed. It incorporates the semi-automated process of building knowledge graphs (KG) optimizing natural LLM understanding. The proposed method combines structured knowledge graph-based search with semantic text retrieval and evaluation to enhance answer quality, reduce hallucination, and improve Q&A performance in the Vietnamese legal domain.

Keywords: Knowledge Base · Knowledge Graph · Retrieval-Augmented Generation · LLM.

1 Introduction

Access to legal information plays a crucial role in supporting legal practice, public governance, and citizens' understanding of their rights and obligations. Legal documents are inherently complex, featuring hierarchical structures, specialized terminology, and dense cross-references between articles and clauses [2]. In Vietnamese legal system, statutes are organized into multiple levels, such as parts, chapters, articles, clauses, and points, creating a highly structured, yet linguistically intricate knowledge space [1,21]. These characteristics pose significant challenges for automated legal information retrieval and question answering systems.

The emergence of Large Language Models (LLMs) has marked a major breakthrough in natural language processing, enabling systems to better understand user intent and generate context-aware responses [3,7]. LLMs demonstrate strong performance

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in open-domain question answering and semantic understanding tasks. Nevertheless, when applied to professional domains in low-resource languages [11,8], these limitations include unreliable legal reasoning, insufficient understanding of structured legal relationships, and a tendency to generate hallucinated or legally inaccurate information [16].

Retrieval-Augmented Generation (RAG) frameworks address these issues by integrating LLM with external knowledge retrieved [17,6]. Although RAG systems improve factual accuracy compared to purely generative models, most existing approaches treat legal texts as unstructured documents and fail to explicitly establish semantic relations within legal knowledge [16]. This results in incomplete contextual grounding and sub-optimal reasoning over interconnected legal provisions.

To address these challenges, this study proposes a domain-specific RAG system tailored to Vietnamese law, explicitly designed to accommodate the linguistic characteristics and structural properties of Vietnamese legal texts. The proposed approach integrates structured legal knowledge with generative language models by combining semi-automated knowledge graph construction, leveraging LLM-based semantic understanding, with a dual-channel retrieval mechanism that unifies knowledge graph-based retrieval and semantic text retrieval. By grounding generation in both structured and unstructured Vietnamese legal sources, the system aims to enhance answer accuracy, reduce hallucination, and improve legal reasoning performance in Vietnamese legal question answering.

The next section reviews related work on legal information retrieval, knowledge graph construction, and RAG in the legal domain. Section 3 describes the knowledge base construction process, including ontology design, data collection, pre-processing pipelines for Vietnamese legal documents and automated knowledge graph extraction. Section 4 and 5 introduce and experiment the hybrid retrieval and reasoning strategy generating grounded legal responses. The last section concludes the results and directions for future work.

2 Related Work

Early intelligent legal systems primarily relied on rule-based and logic-driven approaches, which enabled precise legal reasoning within narrowly defined scopes [12,19]. While effective in controlled settings, these systems were often rigid, domain-specific, and costly to develop and maintain, as they required extensive manual knowledge engineering and struggled to adapt to diverse or evolving legal queries.

To reduce reliance on handcrafted rules, subsequent research shifted toward data-driven language models. Chalkidis et al. propose LEGAL-BERT, a domain-specific pretrained language model trained on large-scale legal corpora [4]. By capturing legal terminology and syntactic patterns more effectively than general-domain BERT, LEGAL-BERT achieves improved performance across various legal NLP tasks. However, it relies solely on implicit textual representations, limiting its ability to support structured legal reasoning and complex question answering.

More recently, RAG has emerged as a promising paradigm for improving the accuracy of LLM-based systems. Lewis et al. introduced a unified framework that enables

language models to dynamically retrieve external non-parametric knowledge during inference [17], grounding generation in relevant documents and substantial improvements on knowledge-intensive tasks. Building on this foundation, Guu et al. [13] integrate retrieval mechanisms directly into the pretraining process, while the research in [14] demonstrates that large-scale retrieval and retrieval-aware fine-tuning can significantly enhance factual grounding and scalability, even rivaling much larger parametric models.

Building on these advances, recent studies have begun to apply RAG-based frameworks to legal information retrieval and legal question answering. By grounding generation in authoritative legal sources such as statutes, case law, and regulatory texts, these systems aim to improve factual accuracy and transparency in legal reasoning [16,5]. Notable examples include LexRAG [18], which explores retrieval-augmented generation in multi-turn legal consultation scenarios, highlighting the practical applicability of RAG-based approaches in real-world legal settings.

In parallel, the complex and hierarchical nature of legal systems has motivated the use of knowledge graphs (KGs) to represent legal knowledge in a structured and relational form [2,22]. Knowledge graphs enable explicit modeling of legal entities, norms, and inter-article relationships, supporting more structured reasoning. The study in [27] utilizes dependency parsing and noun phrase identification to extract subject–relation–object triples from legal texts. These are organized into taxonomies and aligned with legal ontology patterns to enable structured querying, through complex reasoning limitations exist.

3 Knowledge Graph Construction

This section presents the methodology for constructing a Knowledge Graph from Vietnamese legal documents. It details the preprocessing pipeline and the knowledge extraction process that transforms unstructured legal texts into structured knowledge triples, forming the foundation for downstream legal reasoning and question-answering tasks.

3.1 Knowledge base for Vietnamese Legal documents

The Vietnamese legal system follows a tradition of civil law and is characterized by a hierarchical and codified structure. Legal norms are enacted primarily through written legal instruments issued by state authorities, and their validity and application are determined by a well-defined hierarchy, as shown in Table 1. Vietnamese legal texts are organized in a strict hierarchical format, typically decomposed into multiple levels, including Parts, Chapters, Sections, Articles, Clauses, and Points [20].

Building an effective Q&A system in Vietnamese legal domain requires a knowledge representation model capable of capturing both the hierarchical structure of legal texts and the semantic meaning of legal terminology. Legal-Onto is a specialized ontology-based architecture [24]. It is suitable for representing knowledge of current legal documents in the Vietnamese legal system. The structure of Legal-Onto is defined as follows [25,26]:

$$K = (C, R, Rule) \oplus (Concept, Rela) \quad (1)$$

Table 1. Hierarchical Structure of Vietnamese Legal Documents

Level	Issuing Authority	Types of Legal Documents
1	National Assembly	Constitution
2	National Assembly	Codes; Laws; Resolutions
3	Standing Committee of the National Assembly	Ordinances; Resolutions
4	President of the State	Orders; Decisions
5	Government	Decrees; Implementing Resolutions
6	Prime Minister	Decisions; Directives
7	Supreme People’s Court	Council of Judges of the Supreme People’s Court: Resolutions
8	Inter-agency Authorities	Joint Resolutions issued by the Standing Committee of the National Assembly, the Government, and the Presidium of the Vietnam Fatherland Front
9	People’s Councils of all levels	Implementing Resolutions
10	Ministers / Heads of ministerial-level agencies	Circulars

The first component ($C, R, Rule$) represents the structure of the Rela-model [10,26] and serves as the logical backbone of the system. In which, C denotes the set of concepts, where each concept is enriched with an internal structure tailored for organizing legal information, R denotes the set of relations among concepts, keywords, and legal text data sources, and $Rule$ denotes the set of inference rules within the legal knowledge domain.

The second component ($Concept, Rela$) corresponds to a Knowledge Graph that captures relations among legal concepts and keywords in legal texts. In this graph, $Concept$ represents the set of nodes, corresponding to legal concepts or keywords, and $Rela$ represents the set of edges encoding the relations between these nodes.

3.2 LLM-Assisted Legal Text Pre-processing

In this stage, the preprocessing on raw legal text data is described as Figure 1. Legal documents are modeled using a hierarchical structure. Each document is decomposed into logical levels from higher to lower, including Parts, Chapters, Sections, Articles, Clauses, and Points. Representing legal texts preserves semantic context at multiple granularities while facilitating feature extraction, relation identification, and semantic reasoning across textual units.

Based on this hierarchical structure, the texts are normalized and segmented to identify the boundaries of legal units at each level. Large Language Models (LLMs) are integrated as a core semantic processing module within the system architecture to enhance the quality and usability of input text representations [9]. In the legal domain, raw statutory language is often syntactically dense, semantically implicit, and characterized

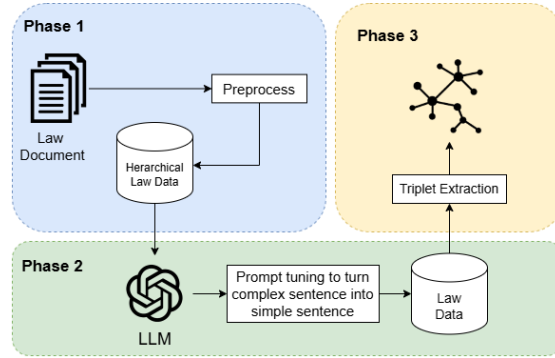


Fig. 1. Data pipeline for hierarchical legal document processing and knowledge extraction.

by long, nested sentence structures. It makes direct application of conventional NLP models for relation and entity extraction highly challenging.

LLM is used as an intermediate semantic layer that bridges unstructured legal language and structured information extraction. It enable subsequent NLP components to perform relation extraction and knowledge triple construction more reliably within each text chunk. This step is critical, as without such semantic normalization, downstream NLP models often fail to accurately identify entities and relations due to implicit legal reasoning patterns.

3.3 Knowledge Graph Construction Process

NLP models are employed to transform normalized legal text into a structured knowledge representation in the form of a Knowledge Graph (KG). The primary objective is to extract legal concepts and their corresponding relations, thereby effectively supporting information retrieval and reasoning tasks. The pipeline for knowledge extraction is shown in Figure 2.

The input to this module consists of sentences that have been simplified and normalized by LLM during earlier pre-processing stages. Each sentence is first tokenized and

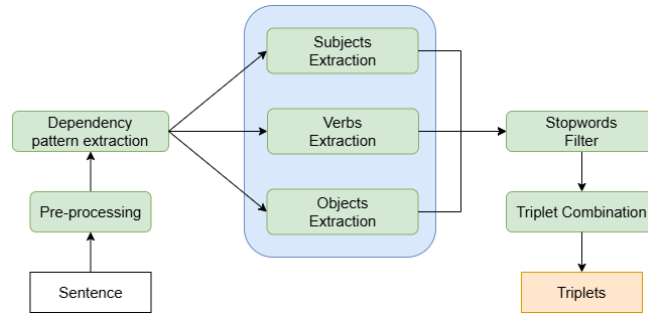


Fig. 2. The knowledge triplet extraction pipeline.

then processed using VNCORENLP [28] model for tokenization and PhoNLP [23] model for part-of-speech tagging and syntactic dependency parsing. The resulting linguistic annotations provide essential syntactic and semantic cues. These components are subsequently normalized and reorganized into a tube (*Subject–Relation–Object*) structure, while preserving the original semantic order of the sentence.

Then the resulting triplets are integrated into the legal knowledge base through an ontology-based mapping process. In this representation, concepts are modeled as nodes and relations as edges within the KG. The constructed knowledge graph serves as a foundational layer for downstream applications such as semantic search, knowledge reasoning, and large-scale legal data analysis.

To maintain consistency with newly issued legal documents, the system adopts an incremental update process. New documents are processed using the same hierarchical segmentation and LLM-based normalization pipeline as in the initial construction. Extracted triplets are aligned with the existing ontology, while amendment and repeal relations are used to update or deactivate outdated nodes and edges. The resulting subgraph is then integrated into the global knowledge graph, and corresponding embeddings are updated in the vector database to ensure consistency between graph-based and semantic retrieval.

4 Legal Information Retrieval System

This section presents the architecture of the proposed legal information retrieval system, which is designed to serve as the retrieval backbone of a retrieval-augmented generation framework. The system integrates a vector database for semantic search with a graph-augmented retrieval mechanism exploiting legal hierarchies and inter-document relationships.

4.1 Graph-Augmented Retrieval

A vector database is employed to manage and index legal documents, thereby enhancing the effectiveness of the Retrieval-Augmented Generation (RAG) pipeline for legal research [15]. The system is designed to efficiently store, compare, and retrieve dense vector representations of legal texts at scale, enabling fast and accurate semantic search over large collections of legal documents.

Legal regulations are segmented into the smallest meaningful legal units while preserving their surrounding contextual information, as described in previous sections. These text segments, along with the textual content associated with nodes in the knowledge graph, are encoded into high-dimensional semantic vector representations and stored in the vector database. After constructing the knowledge graph and vector database from domain-specific legal knowledge source, the system initiates context retrieval based on user queries to support the LLM in generating accurate responses with clear provenance.

To jointly leverage the strengths of semantic retrieval and structure-aware knowledge access, a hybrid retrieval strategy integrates the vector database with the legal knowledge graph. This approach enables the system to capture semantic similarity between

text segments while simultaneously exploiting implicit relationships encoded in knowledge graph triples, as well as links among legal documents. As a result, the proposed retrieval mechanism improves both the accuracy and traceability of retrieval outcomes and generated responses.

4.2 RAG Processing Pipeline

The **query process** is designed as a structured pre-processing stage that transforms natural language queries into standardized and semantically explicit representations to support downstream retrieval. This stage reduces linguistic noise, resolves ambiguity, and extracts the core informational intent. Specifically, user queries are first normalized to standardize abbreviations, legal shorthand, and special characters commonly found in Vietnamese legal texts. They are then refined through spelling and grammatical correction using a LLM, followed by semantic clarification to resolve ambiguous or underspecified expressions without introducing new interpretations. For complex queries, the LLM further decomposes the input into a set of concise sub-queries, each representing a distinct legal issue. Notably, this stage is limited to query restructuring and does not involve answer generation or legal reasoning.

The **retrieval process** integrates knowledge graph-based reasoning with vector-based semantic search to identify contextually relevant legal passages. From the processed sub-queries, keyphrases representing core legal concepts and entities are extracted using a few-shot LLM prompting strategy. These keyphrases are then linked to corresponding nodes in the legal knowledge graph via semantic similarity, forming a set of seed entities. A query-specific subgraph is constructed by expanding these seed nodes with their one-hop neighbors and subsequently refined based on node relevance, relation relevance, and overall semantic alignment with the query. The refined subgraph is used to guide retrieval from a vector database, where candidate passages are retrieved and re-ranked using a hybrid strategy combining BM25 and Dense Passage Retrieval (DPR). Finally, the knowledge graph is leveraged to incorporate amendment, reference, and cross-reference relations among legal provisions, enabling contextual enrichment and improving the completeness and traceability of the retrieved information.

5 Experiments

The legal documents used in this study were collected from official repositories of the Vietnamese government, including the National Database of Legal Documents and the Official Gazette. The dataset consists of 13 statutory and sub-statutory legal documents related to the Land Law domain. All documents are currently in force and were promulgated between 2003 and 2025.

5.1 Knowledge Base Setup

The collected legal texts were structured according to the hierarchical organization of the Vietnamese legal system in Land Law. Table 2 summarizes the number of legal units at each hierarchical level after pre-processing and normalization.

Table 2. Statistics of hierarchical legal document components

No.	Component Type	Quantity
1	Legal documents	13
2	Chapters	125
3	Articles	2,005
4	Clauses	7,009
5	Points	5,634
6	Inter-section relations	2,931

From the structured legal sections dataset, content from leaf-level legal units was collected and processed to extract legal concepts and the relationships between them. Concept and relation extraction was performed automatically using a hybrid pipeline that combines large language models (LLMs) for semantic restructuring with *GPT-4.1-mini* model, VNCORENLP [28] and PhoNLP [23] for part-of-speech (POS) tagging and dependency parsing. The process for constructing the KG of Land Law is presented as Figure 3.

Based on the identified concepts and relations, a knowledge graph was constructed to represent the semantic structure of legal documents, with knowledge encoded in the form of subject–relation–object triplets. The extraction and construction process was fully automated and guided by a domain ontology, ensuring consistency in entity representation and relation typing across documents. The resulting knowledge graph consists of 23,536 nodes, 1,811 edges, and 63,899 triplets.

5.2 Query Results and Evaluation

To evaluate the effectiveness of the proposed system, a comparative experiment is conducted. It uses a dataset of 113 practical legal questions collected from online legal forums and advisory platforms. The questions were categorized into 05 questions types corresponding to different legal topics and levels of complexity, as described in Table 3. Each question was answered independently by three systems: a notebook-based LLM configuration (NotebookLM), ChatGPT 5.1, and the designed Law Assistant chatbot.

Evaluation was performed using exact-answer correctness as the primary evaluation metric, where an answer was considered correct if it accurately identified the relevant

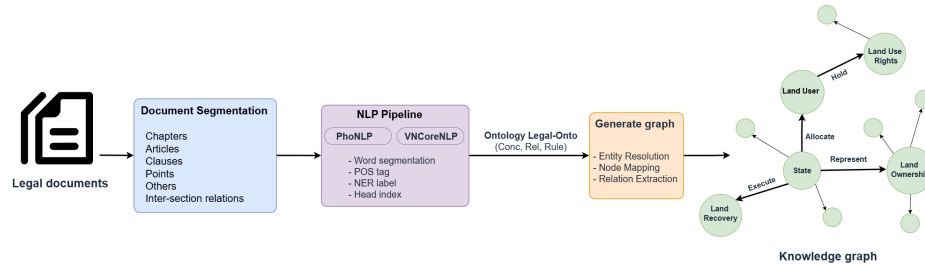
**Fig. 3.** The construction of the KG for Land Law.

Table 3. Classification of land-law questions.

No.	Question Type	Description	Qty
1	Procedural questions	Questions requiring identification of required documents, procedural steps, or implementation workflows for specific administrative procedures.	29
2	General land management	Questions related to land allocation, land lease, land acquisition, compensation, support, resettlement, land registration, and issuance of land use right certificates.	28
3	Land use	Questions concerning land-use purpose conversion, land valuation, land-use planning, and the rights and obligations of land users.	22
4	Specific cases	Civil disputes (e.g., authorization, inheritance, legal representation) and criminal cases (e.g., fraud and land-related disputes).	22
5	Local-specific issues	Questions involving local variations in land price frameworks, transitional legal cases arising before the enactment of new laws, and conflicts in local administrative practices.	12

legal provisions, provided legally valid reasoning consistent with current Vietnamese land law, and passed a predefined IRAC score threshold. The composite IRAC score for each response is calculated using a weighted linear combination of five key legal dimensions as follows:

$$\text{IRAC Score (\%)} = 10I + 35R + 40A + 10C + 5L \quad (2)$$

where, I, R, A, C , and $L \in [0, 1]$ denote the scores assigned by legal experts to each component of the IRAC framework. The weight distribution reflects the relative importance of rule identification and application in professional legal analysis.

Table 4 summarizes the number of correct answers and accuracy rates for each system across different question types.

The results show that the proposed Law Assistant significantly outperforms the baseline systems across most categories, achieving an overall accuracy of **74%**. In contrast, ChatGPT and NotebookLM achieved overall accuracies of **37%** and **35%**, respectively. Notably, the performance gap becomes more pronounced for complex question types, such as those involving land-use rights, dispute resolution, and project-related legal issues (Types 3, 4, and 5), where structured legal knowledge and cross-article reasoning are required.

While ChatGPT demonstrates relatively stable performance on procedural and general management questions (Types 1 and 2), it struggles with questions that require reasoning over hierarchical legal structures or resolving implicit legal relationships, such as amendments and repeals. This limitation becomes more pronounced when new laws or regulatory updates are introduced, as conventional large language models and vector-based RAG systems lack explicit mechanisms to track legal validity, supersession, or temporal dependencies between legal documents. NotebookLM exhibits similar weak-

Table 4. Comparative IRAC evaluation results across 06 question types.

Question Category	Total Count	Model Performance (IRAC Score)		
		NotebookLM	ChatGPT	Law Assistant
Type 1	29	0.33	0.46	0.64
Type 2	28	0.40	0.38	0.87
Type 3	22	0.38	0.10	0.66
Type 4	22	0.39	0.38	0.78
Type 5	12	0.25	0.53	0.73
Overall	113	0.35	0.37	0.74

nesses, particularly in higher-complexity cases, indicating that retrieval and generation pipelines relying solely on unstructured text embeddings are insufficient for maintaining up-to-date legal knowledge and for accurately handling evolving legal frameworks involving amendments and repealed provisions.

Despite strong overall performance, several failure cases were observed, primarily due to limitations in the retrieval process. In some instances, the vector-based retrieval component fails to identify semantically relevant passages, particularly for queries involving rare legal terminology or highly context-dependent interpretations. This leads to incomplete or less relevant context being provided for downstream answer generation.

Nevertheless, the overall results validate the effectiveness of integrating a knowledge graph within a hybrid RAG framework. By explicitly modeling legal concepts and their relationships, the proposed system is better equipped to handle the complexity, structure, and reasoning demands of Vietnamese legal texts, resulting in significantly improved accuracy and robustness in legal question answering.

6 Conclusion and Future work

This paper proposes a hybrid RAG framework for Vietnamese legal question answering that integrates large language models, knowledge graphs, and vector-based semantic retrieval. Unlike conventional RAG systems that rely primarily on unstructured text embeddings, the proposed approach explicitly models legal concepts, hierarchical structures, and inter-article relationships within a knowledge graph, enabling more accurate retrieval and reasoning over legal texts. The experimental results on Vietnamese Land Law show that the proposed method is a promising direction for building robust, explainable, and up-to-date legal AI systems for low-resource and non-English legal domains.

Although the proposed system demonstrates strong performance, the methodological contribution remains primarily at the system integration level. Further investigation is needed to better quantify the individual contributions of each component. In the future,

the combination of ontology-guided knowledge graph traversal with semantic retrieval will be further studied to improve multi-step reasoning capabilities. In addition, the expansion of the legal ontology will incorporate explicit temporal reasoning for tracking legal validity, reduce noise in the triplet extraction process, and scale the framework to additional areas of Vietnamese law.

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